Shortform User Manual Dürkopp 291

Important:
Before this sewing machine is used you need to attend training. It is for your own safety, and the safety of the machine. Spare parts for this machine are expensive or impossible to get, and we hope for the machine to be usable as long as possible.

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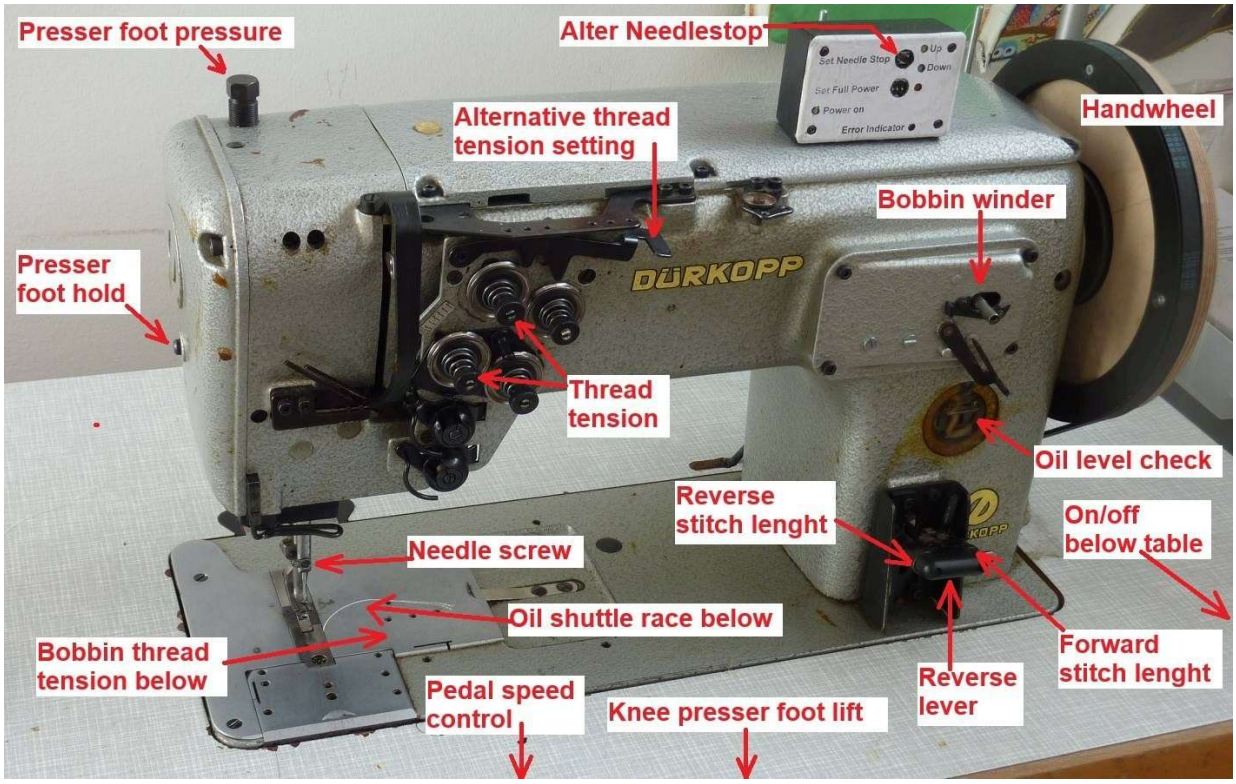
The main controls of the sewing machine



When you switch on power to the machine the computer obtains zero-point measurements of pedal air pressure and motor speed. When you switch on power to the machine await one second before you put pressure on the pedal or turn the handwheel.

Read this before you use the sewing machine

The power and torque from the motor is about ten times higher than you got on a normal household sewing machine. Therefore, the needle is easily pressed through a finger. In case the needle point hits something hard, the needle is likely to be destoyed and needle parts ejected and expensive parts damaged. If the thread gets jammed in the machine, it can be difficult to correct, and some parts can get damaged. Good sewing practices are therefore more important with this kind of machine than with normal sewing machines.

1. Use your hand on the handwheel, when you sew your first stithes on something to get a feeling on how much force is needed. By hand you will normally feel if something is wrong and needs to be corrected.
2. Only turn the machine forward and only backwards max ¼ of a turn if needed.
3. Only turn on power to the machine when needed. It is good for your safety when power is off.
4. Use the start checklist below when you need to use the machine for something.
5. You are only allowed to use the motor on the machine given the conditions provided by the table below. Otherwise, you can damage the machine or get thread jams:

Presser foot	Fabric under presser foot	Needle thread in needle and take-up lever	Bobbin thread in machine	Is machine allowed to work?
Up	Don't care	Yes	Don't care	No
Up	Don't care	Don't care	Yes	No
Down	No	Don't care	Don't care	No
Down	Yes	No	Yes	No
Down	Yes	Yes	No	No
Down	Yes	No	No	Yes
Up	Don't care	No	No	Yes *
Down	Yes	Yes	Yes	Yes **

*Used for bobbin winding
**Used for normal sewing

Start sewing check list

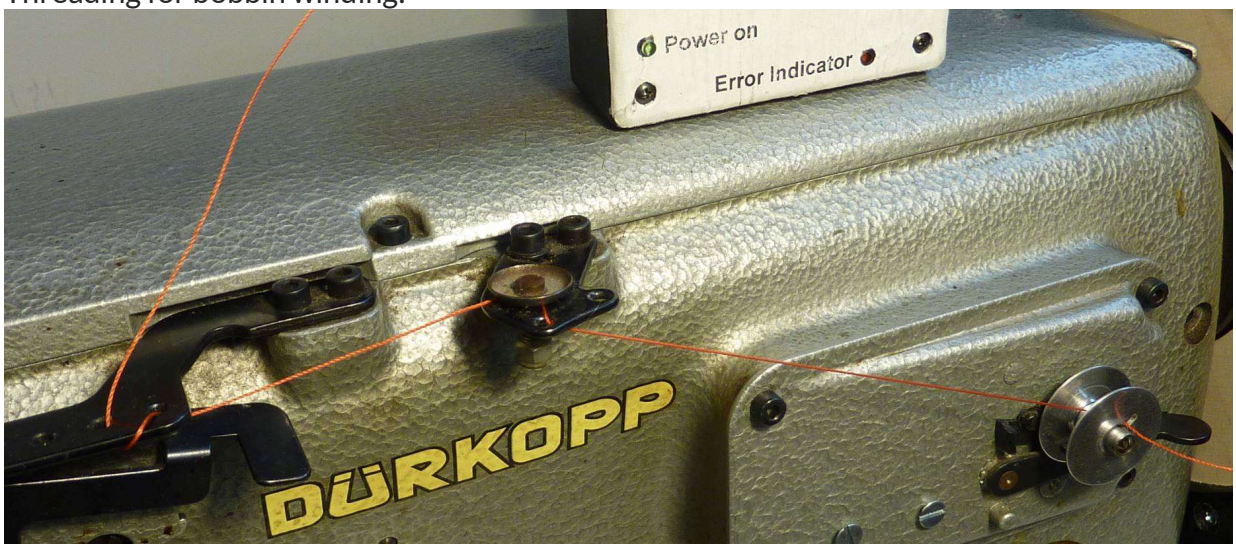
- For safety turn off the sewing machine most of the time
- Remove previous thread from shuttle and from needle to and with take-up lever
- Check oil level on front oil storage and refill to max (only use sewing machine oil)
- Lubricate the shuttle race with two oil drops.
- Raise the presser foot and lock it to an upper position by the black knob to the left on the macchine.
- Remove knee lever and flip the machine to the back and remove belt temporarary.
- Make two turns of the machine by hand. It should feel very easy to turn.
- Check oil level at bottum oil storage and refill to max.
- Provide about 20 drops of oil near buttom suction point.
- Flip machine back. Check belt. Insert knee lever for presser foot lift.
- Deside on a specified lower thread for spool.
- Lift up presserfoot and lock it to an upper position using the black knob to the left on the machine.
- Thread the machine to wind spool – see photo below.
- Turn on power to the sewing machine and make the machine fill an amount of thread on an empty spool.
- Turn off power to the sewing machine.
- Thread upper shuttle as shown in photos below. Feel thread tension by hand. If the thread tension feels wrong, you will check threading and adjust thread tension later.
- Insert the upper shuttle and bobbin into the sewing machine.
- Choose upper sewing thread.
- Choose needle type. Leather point needle for leather. Regular point for other fabrics.
- Choose needle size from table below and make needle size check.
- If the needle has been used before, then you should check its point for damage.
- Insert the needle in the needlebar and turn it right. Use screwdriver, T6-40, that fit the screw for fastning the needle. Tighten firmly the screw by screwdriver.
- Thread the needle thread to the machine until the take-up lever.
- Use spring dynamometer to check thread tension and set it to table value +/- 15%.
- Thread the machine further inclusive needle according to photo.
- Turn the machine by hand to bring up lower thread.
- Use spring dynamometer (1N) to set tension of lower thread according to table value and within +/- 15%.
- Set presser foot pressure to minimum.
- Start test sewing on same or similar fabric. Make first two stitches by hand.
- Turn on the power to sewing machine.
- Make seam trials and adjust upper thread tension to achieve a ballanced stitch
- If the lower thread lays uneven on fabric, you may deside to increase lower thread tension and repeat tests.
- If too high upper tension is needed, you can deside to increase needle size and try again.
- If the presser foot is lifted at any time by the needle friction to fabric, then you will have to increase the down force on presser foot.
- The machine is ready for sewing.

After sewing check list

- Remove thread from needle and eventually cut thread 20 cm above first thread guide.
- Park presser foot on a small piece of scrap fabric.
- Place sticker on machine about thread type and needle size left in machine

Photos of bobbin winding and shuttle threading

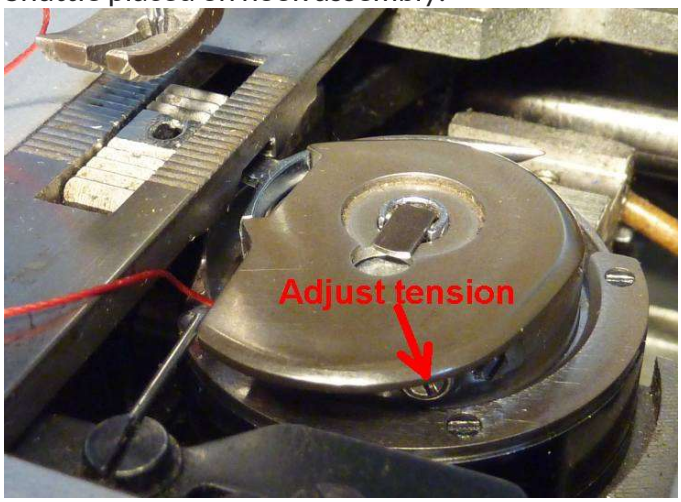
Threading for bobbin winding:



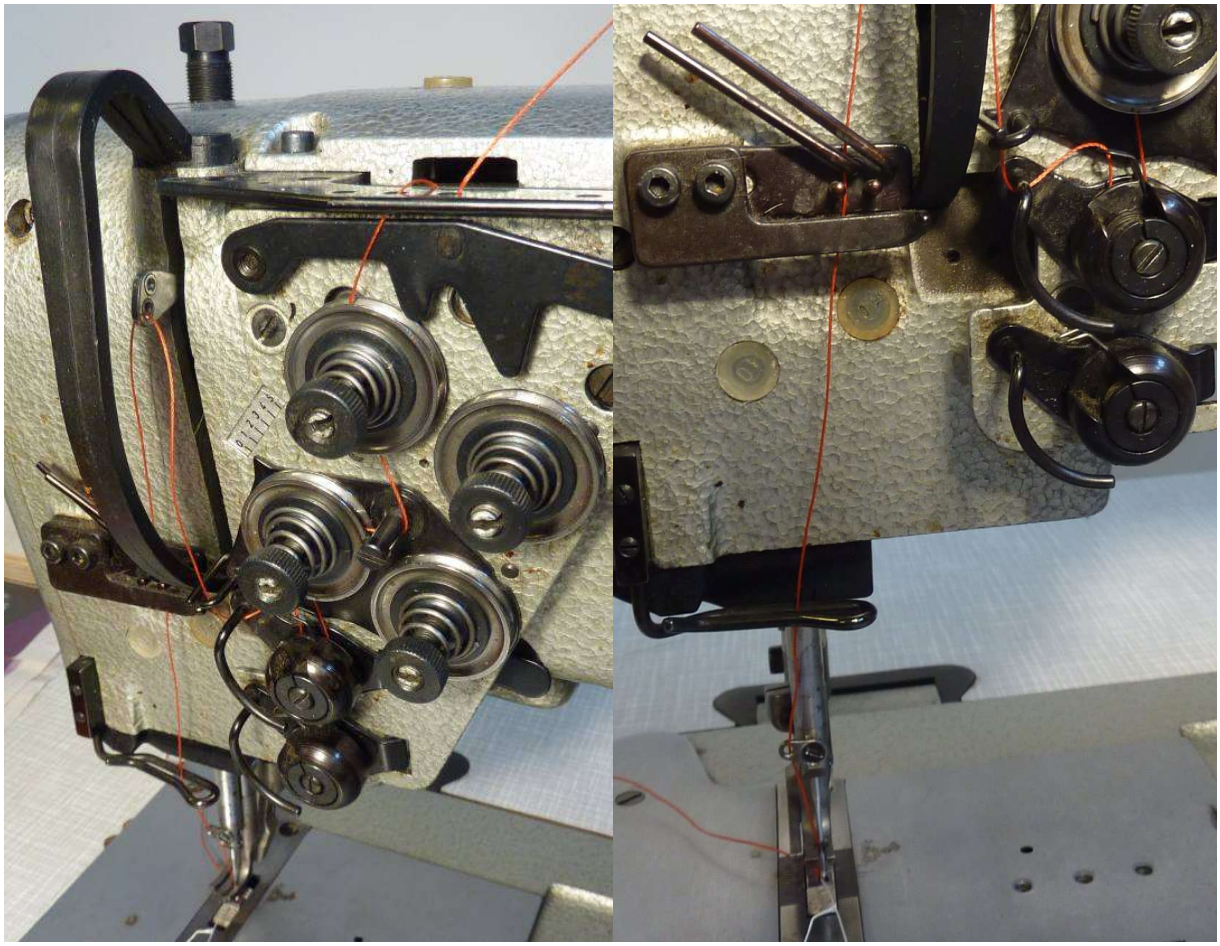
Cut off inside thread on the spool.

Threading of shuttle:

Shuttle placed on hook assembly:



Needle thread threading



Needle grove needs to be on left side of needle. Threading of needle is from left to right. From needle the thread needs to go below through the slit in center presser foot.

You like to have the pre tensioner (first tensioner) make about 1/3 of the total tension, and the main tensioner make about the remaining 2/3 of the total tension.

You measure needle thread tension just before the take-up lever and with no thread in take-up lever.

Recommended initial needle size and thread tensions

This is initial value. Test sewing will likely cause change to these values to achieve the final needle size and tensions. Too low upper tension is risky, and it causes thread jams.

A new needle needs to be turned with the needle groove to the left and firmly fastened by be a correct screwdriver (T6-40) in needle screw.

The thread tension is measured by spring dynamometers.

Fabric	Thread	Needle size	Bobbin thread tension [N]	Needle thread tension [N]
4 layers normal fabric	Serafil 40	100 R	0.35 (max 0.6)	1.7
2 layers tarpauline	Serafil 40	110 R	0.35 (max 0.6)	4
3 layers PP webbing	Serafil 40	110 R	0.35 (max 0.6)	4
2 layers 1.5 mm soft leather	Serafil 40	100 LR	0.35 (max 0.6)	3
2 layers of 2.2 mm firm leather	Serafil 40	120 LR	0.35 (max 0.6)	4
3 layers PP webbing	Serafil 20	130 R	0.5 (max 1.0)	5
3 layers PE firm webbing	Serafil 20	140 R	0.5 (max 1.0)	8
2 layers of 1.5 mm soft leather	Serafil 20	130 LR	0.5 (max 1.0)	5
2 layers of 2.2 mm firm leather	Serafil 20	140 LR	0.5 (max 1.0)	6
4 layers of 2.3 mm firm leather	Serafil 20	140 LR	0.5 (max 1.0)	8
2 layers of tarpauline	Serafil 20	140 R	0.5 (max 1.0)	7
6 layers of tarpauline	Serafil 20	140 R	0.5 (max 1.0)	12
2 layers of 3 mm hard leather	Serafil 20	140 R	0.5 (max 1.0)	9

Error Indicator

An Arduino microcontroller is used to control the sewing machine motor speed. Several checks of the hardware are done by software in this microcontroller. For normal operation the LED should not flash.

If a failure is detected by the software the error indicatior LED will flash and the motor stops. The LED repeats a flashing and then pauses flashing and repeat again. You should count the number flashed and report it to Fablab.

This is what the number means:

1. Warning temperature power electronics or motor.
2. Shut down due to high temperature power electronics.
3. Shut down due to high temperature of motor windings.
4. The sewing machine got blocked or you got serious hardware issue.
5. Error in needle stop.
6. Speed Pedal pressure incorrect at startup.
7. 5 V power supply outside voltage specification.
8. 24 V power supply outside voltage specification.
9. Startup of machine while it is not stopped.
10. Failure in protection of regenerative breaking of sewing machine.